

Goal

Run the following panel event study:

$$Y_{ct} = \alpha_c + \lambda_t + \sum_k \beta_k (\text{Exposure}_c \times 1[t = k]) + (X_c \times \lambda_t)' \Gamma + \varepsilon_{ct}$$

where, given a commuting zone (CZ) c and year t , we have:

- Y_{ct} , Republican margin in presidential election year t
- α_c , CZ fixed effects
- λ_t , election-year fixed effects
- Exposure_c , the ADH measure of 1990-2007 exposure
- X_c , a column vector of CZ-level baseline controls
 - 1990 manufacturing share, college share, and foreign-born share

We focus on running the event study so we can study separate horizons:

- Short-run effects (say, 2008-2012) are likely influenced by the recession and Obama-era response
- Medium-run effects (say, ~2016) are likely influenced by Trump-specific rhetoric
- Long-run effects (e.g., 2020+) are likely influenced by COVID, inflation, etc.

Process

First:

We begin by building a map from current-day counties (or at least 2024 counties; there have been no substantial changes to counties since 2022, anyways) to 2023 counties to 2022 counties etc., all the way to 1990 counties and then to 1990 CZs. The process is like so:

1. Map 2020 counties to 1990 counties using [Ferrara, Testa, and Zhou \(2024\)](#).

1. Weighting rules:

1. Area-based (model 1, or M1).
2. Population-based (M2), with county area divided into urban and rural areas.
3. Population-based (M3), with county area divided into urban and rural areas after excluding non-inhabitable areas.
4. Population-based (M4), with county area divided into urban and rural areas after excluding non-inhabitable areas, with additional weighting for topographic suitability (i.e., elevation).
5. **Population-based (M5), with built-up settlement areas indicated in space (1810–2020 only).**

6. Population-based (M6), with built-up property counts indicated in space (1810–2020 only).
2. Aggregate 1990 counties to 1990 CZs using `replication/cz-data`, sourced from <https://www.ers.usda.gov/data-products/commuting-zones-and-labor-market-areas>
 1. Nothing complicated here. `cz-198090.xls` should match county names and FIPS codes to 1990 CZs. As far as I could tell, the county names and codes are from 1990 and the CZ codes match Tolbert and Sizer (1996), the paper ADH 2013 used for their CZs.
3. Match with the data from the replication package, `replication/adh2013`
 1. Everything's smooth sailing now; just join along the 1990 CZ code. Also don't forget that out of 741 CZs, ADH only use "the 722 CZs that cover the entire mainland United States (both metropolitan and rural areas)" (p. 2132).
4. Check and address errors, e.g.:
 1. Inconsistencies in FIPS codes
 2. Missing counties or duplicated mappings
 3. Population weights which don't sum to 1

Second, use this to match adh2013 data with variables for the political outcome Y_{ct}

- Data sources of Y_{ct}
 - Use `replication/outcomes-data/aa2021-nospatial`, sourced from [Replication Data for: Partisanship & Nationalization in American Elections: Evidence from Presidential, Senatorial, & Gubernatorial Elections in the U.S. Counties, 1872-2020](#), the replication package of Amlani and Algara (2021), to construct 1950-2020 political outcomes in presidential, senatorial, and gubernatorial elections. We will use this for pre-trends before 1990, trends during the 1990-2007 shock, and post-shock outcomes. This will be our primary source of outcomes.
- For now, define Y_{ct} as the Republican presidential vote margin

Third, prepare Conley spatial HAC SEs.

- Get county centroids from `replication/ftz2024/crosswalks/County-CD-centroid-lat-lon/lat_lon_coordinates_county_csv/counties_1990_xy.csv`.
- Map counties to 1990 CZs
- Compute population-weighted CZ centroids using 1990 county populations in `replication/cz-data/cz-198090.xls`, sourced from <https://www.ers.usda.gov/data-products/commuting-zones-and-labor-market-areas>

$$- \text{ CZ centroid} = \sum_{i \in c} w_i \cdot (\text{lat}_i, \text{lon}_i), \quad w_i = \frac{\text{pop}_{i,1990}}{\sum_{j \in c} \text{pop}_{j,1990}}$$

- Calculate [Conley \(1999\)](#) spatial HAC SEs. For reference, recall:

Homoskedastic:

$$SE(\hat{\beta}) = \sqrt{\frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2}}$$

Heteroskedastic (HC0):

$$SE(\hat{\beta}) = \sqrt{\frac{\sum (x_i - \bar{x})^2 \hat{\varepsilon}_i^2}{[\sum (x_i - \bar{x})^2]^2}}$$

Cluster-robust:

$$SE(\hat{\beta}) = \sqrt{\frac{\sum_g [\sum_{i \in g} (x_i - \bar{x}) \hat{\varepsilon}_i]^2}{[\sum (x_i - \bar{x})^2]^2}}$$

Conley:

$$SE(\hat{\beta}) = \sqrt{\frac{\sum_i \sum_j K(d_{ij})(x_i - \bar{x})(x_j - \bar{x}) \hat{\varepsilon}_i \hat{\varepsilon}_j}{[\sum (x_i - \bar{x})^2]^2}}$$

Fourth, run the event study

- Note that potential controls X_{ct} from 1990 can be found in the ADH 2013 replication package.

Fifth, implement future plans:

Check for errors, attenuation, missingness, etc.

- If I recall correctly, extending out to 1950 induces near-singularity
 - Why could this be happening?
- Why such absurdly low vote-retention rates?
 - `retained_vote_share` is computed after the many-to-many crosswalk join, so the denominator double- or multi-counts source-county votes when a county has multiple crosswalk rows?
 - Repair matching algorithm
- Conley VCOV (variance-covariance matrix) was not positive definite, had to be repaired (i.e., numerically regularized, I suppose?)

- Is `vcov_fix = TRUE` ok?
 - Inspect role of interacted controls in near-singularity
- Is the panel fully balanced?

Pre-trend

- Inspect pre-trends of the outcomes variables in `replication/outcomes-data/aa2021-nospatial`

Robustness checks

- Different SEs
 - Different Conley SEs
 - * Test out lower Conley cutoffs. Test for bandwidth sensitivity
 - * Perhaps use the `SpatialInference` R package for the bandwidth selection method described in [Lehner \(2026\)](#).
 - Different types of SEs
 - * Compare Conley SEs with heteroskedastic, CZ-clustered, state-clustered, and perhaps even alternative spatial SEs.
- Run specifications with/without instrument and run weak instrument tests (medium-priority)
 - Exposure_c, the ADH measure of exposure, is an endogenous regressor!!
 - * ADH's Bartik instrument for their exposure measure is Chinese import growth in other high-income countries. Will likely be far weaker for our longer-run political outcomes than their shorter-run labor outcomes.
 - May need to use the spatial HAC SEs of [Kelejian and Prucha \(2006\)](#)
 - How to test?
 - * One instrument → first-stage F -stat is natural, test if industry shares predict exposure
 - * But it's a Bartik instrument! [Goldsmith-Pinkham, Sorkin, and Swift \(2020\)](#), a single industry share likely correlated with the error term may drive this. Might have to do some Rotemberg weight funkiness. [Borusyak, Hull, and Jaravel \(2025\)](#) could be a helpful guide here too.
- Test for Bartik instrument flaws (low)
 - Presence of flaws can be tested by replicating Sec. A.5 of the [Online Appendix](#) for [Goldsmith-Pinkham, Sorkin, and Swift \(2020\)](#).
 - Additional sensitivity can be tested using [Apfel \(2024\)](#)

- Test for sensitivity to commuting zone definition
 - Commuting zone definitions are sensitive, making empirical results sensitive
 - Replicate [Foote, Kutzbach, and Vilhuber \(2021\)](#) using <https://github.com/larsvilhuber/MobZ> i.e., <https://larsvilhuber.github.io/MobZ/> or <https://zenodo.org/records/4072428>
- Test for crosswalk flaws
 - Compare the sensitivity of results to use of different weights (e.g., M5 vs. M6) in step 1 of crosswalk.
 - Any way to check attenuation from crosswalk?
 - Check robustness of results to other crosswalks e.g., from <https://www.ddorn.net/data.htm>
- Address the flaws of ADH 2013 data
 - Ex: Sectoral/product aggregation bias (see [Wang et al., 2018](#)), temporal aggregation bias (see [Rothwell, 2017](#)), inconsistent industry coding and missing firm reorganization (see [Bloom et al., 2024](#)).
 - Reconstruct exposure measure
 - * E.g., Replace static exposure with time-varying exposure.
 - Single static 1990-2007 exposure with election-year dummies could work better if we instead used time-varying exposure by subperiod, namely 1991-1999 and 2000-2007
 - Relevant data from <https://www.ddorn.net/data.htm> to reconstruct a better exposure
 - * All files from [C] Industry Codes and [D] Industry Trade Exposure

More outcomes and mechanisms

- Run specifications where Y_{ct} , a CZ-level political outcome in year t is either Republican vote share, Democratic vote share, two-party share, Republican margin, and swings across elections.
- Robustness to different data source for electoral outcomes
 - Use `replication/outcomes-data/pres_elections-2000-2024`, sourced from [County Presidential Election Returns 2000-2024](#), for county-level political outcomes in presidential elections from 2000-2024.
 - * Our “First” step in the Process would then also have to map 2024 counties to 2020 counties using [Lutz \(2025\)](#).
 - The only [substantial changes to counties](#) made this period were in Connecticut in 2022. [Lutz \(2025\)](#) provides a county-level crosswalk of this. Directly replicating the code in the README of `replication/lutz2025/geolinkr-main`, sourced from <https://github.com/ChandlerLutz/geolinkr>, is sufficient for this. Note how his code reads in external data via URLs.

- **Weighting rule:** The README says he uses “the number of housing units in each census tract as weights.”
- Build outcome variable for election turnout
 - [National Neighborhood Data Archive \(NaNDA\): Voter Registration, Turnout, and Partisanship by County, United States, 2004-2022 \(ICPSR 38506\)](#) is very promising
 - * Relevant county-level variables: Registered voters, ballots cast, voting population (specifically CVAP), voter registration, voter turnout, registered voter turnout
- Build outcomes that provide a better picture of the full political effect
 - Motivation: [Autor et al. \(2020\)](#) find the China shock’s main political effects to be a rightward shift in media consumption and electoral outcomes + a loss of moderates.
 - Partisanship
 - * [National Neighborhood Data Archive \(NaNDA\): Voter Registration, Turnout, and Partisanship by County, United States, 2004-2022 \(ICPSR 38506\)](#) is very promising
 - Relevant county-level variables: Democratic and Republican partisanship indices
 - Polarization
 - * [American National Election Studies \(ANES\)](#) data measures polarization and other political attitudes; read the questions it asks [here](#). Sample sizes per CZ are probably tiny, however. Don’t prioritize it.
 - Data on media consumption?
 - Relevant political outcomes from <https://www.ddorn.net/data.htm>:
 - * [\[H1\]](#) Liberal, moderate and conservative legislators in U.S. Congress, 2002-2016.
 - * [\[H2\]](#) Members of the Tea Party, Liberty and Freedom Caucuses, 2010-2016
- Test mechanism, which should be very hard to genuinely determine. Could be any one of:
 - Displaced manufacturing workers becoming more protectionist, conservative, Republican, etc.
 - Migration
 - * Could use [SOI tax data](#) to allow migration as a mediator
 - CZ composition changes
 - * Could find mediators from the Census Bureau’s [PEP](#) or [ACS](#), the NCI’s [SEER](#), IPUMS [NHGIS time series](#), or [other public data sources](#)
 - * Section [E] Local Labor Market Geography of <https://www.ddorn.net/data.htm> could be useful for merging these
 - Workers persuading each other
 - Elite rhetoric
 - * Could try and reconstruct measures from [Autor et al. \(2020\)](#) using publicly-available data

- Changes in turnout
- Better party sorting of voters
- Relevant data from <https://www.ddorn.net/data.htm>, e.g., for baseline manufacturing mix or tradable exposure
 - * [F1] CBP county-level employment by industry, 1988, 1991, 1999, 2007 and 2011.
- Did a political change occur for majority-white + trade-exposed CZs vs. majority-white + not trade-exposed CZs? Or did it only occur for majority-white CZs vs. majority-minority CZs?
- Does media consumption (e.g., Fox News viewership) predict political change?

Pretty stuff for the paper

- Make maps
 - 1990 census tract shapefiles in `replication/spatial-data/Census_Tracts_in_1990`, sourced from <https://catalog.data.gov/dataset/census-tracts-in-1990>.
 - 1990 county shapefiles at `replication/spatial-data/counties-1990`, sourced from <https://geo.btaa.org/catalog/stanford-pb817xw6983>

Additional literature review

- https://en.wikipedia.org/wiki/China_shock
- [Partisan Popularity & Protectionism: The Political Economy Impact of the China Shock on Local Labor Markets](#)